

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

QUESTION 5 – DEPTH ESTIMATION STRATEGY

Problem Statement

A depth estimation system has the following constraints:

- Only **one camera** is available.
- **Moderate depth accuracy** is acceptable.

The available techniques are:

- **Stereo Vision:** High accuracy, but requires two cameras.
- **Structure from Motion:** Moderate accuracy and requires only one camera.

Analysis

Stereo Vision estimates depth using information from two camera viewpoints. Although it provides high accuracy, it cannot be used in this situation because the system has only **one camera**.

Structure from Motion (SfM) can estimate depth using a **single camera** by analyzing images from different viewpoints or frames. The method provides moderate accuracy, which satisfies the requirement of the problem.

Comparison

Technique	Accuracy	Camera Requirement	Suitability
Stereo Vision	High	Two cameras	Not suitable
Structure from Motion	Moderate	One camera	Suitable

Decision

Structure from Motion (SfM) should be selected.

Justification

Structure from Motion satisfies both constraints of the problem. It requires only **one camera**, and its **moderate depth accuracy** is acceptable. Stereo Vision cannot be selected because it requires two cameras.

Final Answer: Structure from Motion (SfM)

QUESTION 6 – RETAIL MOTION TRACKING LOGIC

Problem Statement

A retail system requires:

- Accurate tracking when **crowd density is low**.
- Stable tracking when **crowd density is high**.

The observations are:

- **Optical Flow:** Accurate in low-density scenes but noisy in high-density scenes.
- **Motion Models:** Stable in high-density scenes but less detailed.

Analysis

Optical Flow is suitable when the crowd density is low because it provides accurate and detailed information about movement. However, in high-density situations, multiple people moving simultaneously can make Optical Flow noisy.

Motion Models are more suitable for high-density scenes because they provide stable tracking. However, they are less detailed than Optical Flow.

Therefore, neither technique alone completely satisfies both requirements.

Recommended Approach

An **adaptive combination of Optical Flow and Motion Models** should be used:

- **Low crowd density** → **Optical Flow**
- **High crowd density** → **Motion Models**

Comparison

Crowd Condition	Optical Flow	Motion Models	Preferred Method
Low density	Accurate	Less detailed	Optical Flow
High density	Noisy	Stable	Motion Models

Decision

A combination of Optical Flow and Motion Models should be used.

Justification

The combination allows the system to use the most appropriate technique according to crowd density. Optical Flow provides accurate tracking in low-density scenes, while Motion Models provide stable tracking in high-density scenes.

Final Answer: Combination of Optical Flow and Motion Models

QUESTION 7 – DEFECT DETECTION SYSTEM DECISION

Problem Statement

A defect detection system must:

- Detect subtle defects with **more than 95% accuracy**.
- Adapt to **product variations**.

The available approaches are:

- **Traditional Methods:** Fast, but less adaptive.
- **Deep Learning:** Accurate, but requires training.

Analysis

Traditional computer vision methods are generally fast, but they are less adaptive when product characteristics change. Variations in product appearance can make it difficult for traditional methods to identify all defects reliably.

Deep Learning can learn complex visual features and patterns from training data. This makes it more suitable for detecting subtle defects and handling product variations.

Although Deep Learning requires training, its ability to learn complex patterns makes it more appropriate for the given requirements.

Comparison

Requirement	Traditional Methods	Deep Learning
Detect subtle defects	Less suitable	More suitable
High accuracy	Limited	Suitable
Adapt to product variations	Less adaptive	More adaptive
Training requirement	Lower	Requires training

Decision

Deep Learning should be selected.

Justification

Deep Learning is the most suitable approach because the system requires **very high defect-detection accuracy** and must adapt to **product variations**. Traditional methods are faster but less adaptive. Therefore, Deep Learning provides a better solution for the stated requirements.

Final Answer: Deep Learning

QUESTION 11 – TRAFFIC SURVEILLANCE MODEL SELECTION

Problem Statement

A traffic monitoring system must satisfy the following conditions:

- Accuracy $\geq 90\%$
- Latency ≤ 80 ms
- Works under varying lighting conditions

Given Methods

Method	Accuracy	Latency
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Method A	88%	60 ms
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Method B	91%	85 ms
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Method C	92%	75 ms
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Analysis of Method A

Method A provides **88% accuracy** and **60 ms latency**.

Although the latency satisfies the requirement, the accuracy is below 90%.

Result: Not suitable.

Analysis of Method B

Method B provides **91% accuracy** and **85 ms latency**.

The accuracy requirement is satisfied, but the latency exceeds the maximum allowed 80 ms.

Result: Not suitable.

Analysis of Method C

Method C provides **92% accuracy** and **75 ms latency**.

Both values satisfy the specified requirements.

Result: Suitable.

Comparison

Method	Accuracy Requirement	Latency Requirement	Decision
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Method A	Not satisfied	Satisfied	Reject
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Method B	Satisfied	Not satisfied	Reject
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Method C	Satisfied	Satisfied	Select
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Decision

Method C should be selected.

Justification

Method C is the only method that satisfies both numerical constraints. It provides **92% accuracy**, which is greater than the required 90%, and **75 ms latency**, which is below the maximum limit of 80 ms. Therefore, Method C is the most suitable method.

For the varying-lighting requirement, suitable illumination-robust processing should also be considered during implementation.

Final Answer: Method C

QUESTION 14 – MOTION TRACKING IN SPORTS ANALYTICS

Problem Statement

A sports analytics system requires:

- Accurate motion tracking.
- Real-time performance.

The available techniques are:

- **Optical Flow:** Accurate, but slow.
- **Motion Estimation:** Fast, but less accurate.

Analysis

Optical Flow provides accurate motion tracking and detailed motion information. However, it is slow, which creates a problem because the system needs to operate in real time.

Motion Estimation provides faster processing and is therefore better suited for real-time applications. Although it is less accurate than Optical Flow, its speed makes it more practical for real-time operation.

Comparison

Feature	Optical Flow	Motion Estimation
Accuracy	High	Lower
Speed	Slow	Fast
Real-time suitability	Lower	Higher
Overall suitability	Less suitable	More suitable

Decision

Motion Estimation should be selected.

Justification

Motion Estimation provides the processing speed required for real-time sports analytics. Optical Flow provides better accuracy but is slow. Therefore, Motion Estimation is the more suitable choice when real-time operation is required.

Final Answer: Motion Estimation

QUESTION 24 – RETAIL ANALYTICS SYSTEM

Problem Statement

A retail analytics system requires:

- Customer tracking.
- Moderate accuracy.
- Real-time processing.

The available methods are:

- **Method A:** Accurate, but slow.
- **Method B:** Moderate accuracy, but fast.

Analysis

Method A provides accurate tracking, but its slow processing makes it unsuitable for a system that requires real-time operation.

Method B provides moderate accuracy and fast processing. Since the problem states that moderate accuracy is acceptable, Method B satisfies the accuracy requirement while also supporting real-time processing.

Comparison

Requirement	Method A	Method B
Customer tracking	Suitable	Suitable
Moderate accuracy	Suitable	Suitable
Real-time processing	Not suitable	Suitable
Overall suitability	Lower	Higher

Decision

Method B should be selected.

Justification

Method B provides a suitable balance between accuracy and processing speed. It meets the requirement for moderate accuracy and supports real-time processing. Therefore, Method B is more appropriate than Method A.

Final Answer: Method B

QUESTION 25 – OBJECT DETECTION IN FOGGY CONDITIONS

Problem Statement

An object detection system must:

- Be robust under poor visibility.
- Achieve accuracy $\geq 85\%$.

Given Models

Model	Accuracy	Processing
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Model A	80%	—
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Model B	87%	—
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Model C	90%	Slow
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Analysis of Model A

Model A provides **80% accuracy**, which is below the required 85%.

Result: Reject.

Analysis of Model B

Model B provides **87% accuracy**, which is above the required 85%.

Result: Suitable.

Analysis of Model C

Model C provides the highest accuracy of **90%**, but it is slow. Although it satisfies the accuracy requirement, slow processing may reduce its practical suitability.

Comparison

Model	Accuracy $\geq 85\%$	Speed	Decision
Model A	No	—	Reject
Model B	Yes	Not specified as slow	Select
Model C	Yes	Slow	Less suitable

Decision

Model B should be selected.

Justification

Model B achieves **87% accuracy**, which satisfies the minimum accuracy requirement. Model A fails to reach the required accuracy, while Model C is slower despite having higher accuracy. Therefore, Model B provides the most suitable balance based on the stated constraints.

Final Answer: Model B

QUESTION 28 – SMART HOME SECURITY

Problem Statement

A smart home security system must:

- Detect intrusions accurately.
- Have low computational requirements.

The available options are:

- **YOLO:** Accurate.
- **Viola-Jones:** Efficient.

Analysis

YOLO is an accurate object detection approach. However, the problem specifically states that the system should have **low computational requirements**.

Viola-Jones is an efficient detection technique and is more appropriate when computational resources are limited.

Therefore, the decision should prioritize computational efficiency while still providing the required intrusion detection capability.

Comparison

Feature	YOLO	Viola-Jones
Detection capability	Accurate	Suitable
Computational efficiency	Lower	Higher
Suitable for low computation	Less suitable	More suitable
Overall choice	Reject	Select

Decision

Viola-Jones should be selected.

Justification

Viola-Jones is the more suitable option because the system has a **low-computation constraint**. Although YOLO is accurate, the problem emphasizes computational efficiency. Therefore, Viola-Jones is the appropriate choice under the stated conditions.

Final Answer: Viola-Jones